

Voltage (V)

↳ Difference in electric potential

$$V = \frac{dw}{dq} = \Delta V = V_A - V_B$$

Rise: 4V
 $\frac{dV}{dt} = 4V$

Charge (Q or q)

↳ electrical property of particles

$$q_{e^-} = 1.602 \times 10^{-19} \text{ C}$$

$$q = \int_{t_0}^{t_1} i \, dt$$

Current (i)

↳ Rate of change of charge

$$i = \frac{dq}{dt}$$

Power (P)

$$P = VI = I^2 R = \frac{V^2}{R}$$

↳ Positive: absorbing energy
↳ Negative: delivering energy

$$E = \int_{t_0}^{t_1} p \, dt$$

Tellegen's Theorem / Conservation of energy

Power supplied = Power absorbed

$$\sum p = 0$$

Resistance, resistors (R)

$$R = \frac{\rho L}{A}$$

Random fact
 $1 \text{ kW} = 3.6 \text{ MJ}$

Equivalent R or G

Series:

$$R_{eq} = R_1 + R_2 + R_3 + \dots$$

$$G_{eq} = \left(\frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3} + \dots \right)^{-1}$$

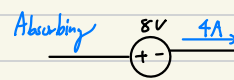
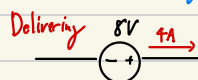
Parallel:

$$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots \right)^{-1}$$

$$G_{eq} = G_1 + G_2 + G_3 + \dots$$

Passive sign convention

↳ +ve current out of +ve side voltage → delivering
absorbing otherwise



Conductance, G (Siemens σ)

$$G = \frac{1}{R} = \frac{A}{V} = \frac{\sigma A}{L}$$

$$\text{Conductivity, } \sigma = \frac{1}{\rho}$$

$$G = \frac{I}{V}, P = \frac{I^2}{G} = V^2 G$$

Parallel / Series circuit

Series:

$$i = i_1 = i_2 = i_3 + \dots$$

$$V = V_1 + V_2 + V_3 + \dots$$

Parallel:

$$i = i_1 + i_2 + i_3 + \dots$$

$$V = V_1 = V_2 = V_3 + \dots$$

Node, branches, loops

Branches, b : single element

Node, n : connection b/w ≥ 2 branches

Loops, l : closed path in circuit

Fundamental theorem of network topology
 $b = l + n - 1$, $l \rightarrow \#$ of independent loops

Parallel current division

$$i_n = \frac{G_n}{G_1 + G_2 + G_3 + \dots} (i)$$

Linearity

$k(iR) = kV$
↳ Applies to linear circuits, any components.

Series voltage division

$$V_n = \frac{R_n}{R_1 + R_2 + R_3 + \dots R_n} (v)$$

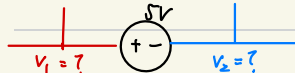
Superposition

↳ Short-cct voltage source + open-cct current source,
always keep dependent unchanged

Nodal analysis

$$i = (v_1 - v_2) / R$$

↳ Supernode, occurs when a voltage source is connected b/w two non-reference node ($\neq 0V$ or known value voltage)

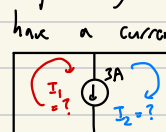


↳ To solve: 1. Find constraint eq. $v_1 = v_2 + 5V$
2. Apply KCL or KVL w/o the source

Mesh analysis

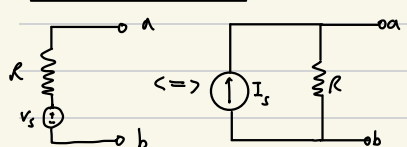
$$V = iR$$

↳ Supermesh, occurs when 2 meshes have a current source in common



↳ To solve: 1. Find constraint eq. $i_1 + 3A = -i_2$
2. Use the whole mesh

Source transformation



Thevenin / Norton Equivalent

R_{TH} :

↳ Only indep. source: Turn all sources off, find R_{eq} .

↳ Only dep. source: Apply test voltage (or current), find $R = \frac{V}{I} \rightarrow V_{TH}$ or $I_{sc} = 0$

↳ Both dep, indep source: Don't turn off anything, find V_{oc} , I_{sc} , $R_{TH} = \frac{V_{oc}}{I_{sc}}$

Key features

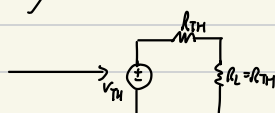
↳ $R_{TH} = R_{eq}$

↳ $V_{TH} = R_{TH} (I_{sc})$

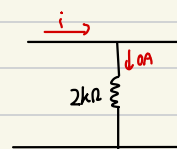
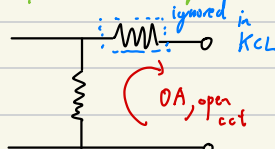
Maximum Power Transfer

$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{I_{sc}^2 R_{TH}}{4}, \text{ when}$$

Load resistor = R_{TH}



Tips on finding V_{TH} :



All current passes through short cct.

Maximum pulse:



Op-Amps

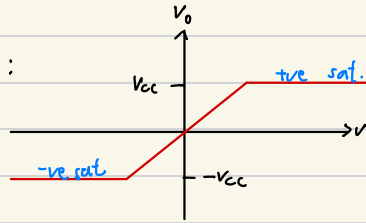
o Ideal Op-Amp:

↳ Input $i_1, i_2 = 0A$

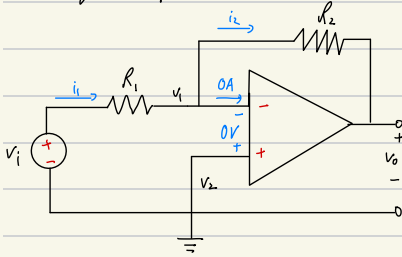
↳ Equal input $v_1 = v_2$

↳ $A_0 = \infty$

↳ Saturation:



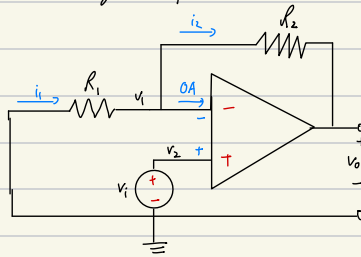
o Inverting Amplifier:



$$v_o = -\frac{R_2}{R_1} v_i, \text{ Voltage gain} = -\frac{R_2}{R_1}$$

↳ Reverses polarity, and amplifies it

o Non-inverting Amplifier

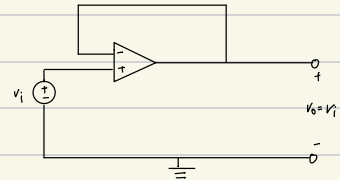


$$v_o = \left(1 + \frac{R_2}{R_1}\right) v_i, \text{ Voltage gain} = 1 + \frac{R_2}{R_1}$$

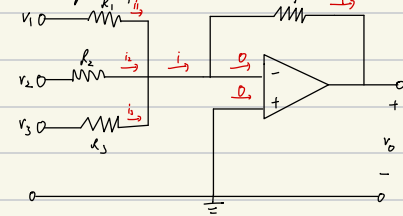
↳ Provide the voltage gain, never < 0 , $\frac{R_2}{R_1} \geq 0$

o Unity-gain buffer amplifier / voltage follower

↳ Useful for isolating one circuit from another



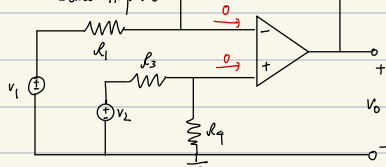
o Summing Amplifier



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3\right)$$

The output is a "weighted sum" of inputs

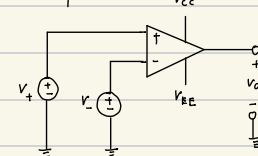
o Difference Amplifier



$$v_o = \frac{R_2}{R_1} \left(\frac{R_1}{R_1 + R_2} v_2 - \frac{R_2}{R_1 + R_2} v_1 \right) \rightarrow \text{If } R_1 = R_2, R_3 = R_4$$

the output is the difference of voltage sources

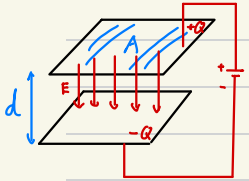
o Comparators



$$v_o = \begin{cases} V_{cc} & v_i > V_{ref} \\ V_{ee} & v_i < V_{ref} \end{cases}$$

Capacitor, capacitance

↳ Parallel-Plate Capacitor



o E, electric field = $\frac{v_o}{d} = \frac{Q}{A \epsilon_r \epsilon_0}$

o Capacitance, $C = \frac{Q}{v_o} = \frac{A \epsilon_r \epsilon_0}{d} \left[F = \frac{C}{V} \right]$

↳ ϵ_r = relative permittivity
 ϵ_0 = permittivity of free space $\sim 8.85 \times 10^{-12} \text{ [F/m]}$

o Current

$$i(t) = C \frac{dv(t)}{dt}$$

o Power

$$p(t) = v(t) i(t)$$

* discharge $\rightarrow$ -ve
absorb $\rightarrow$ +ve

o Voltage

$$v(t) = v_o + \frac{1}{C} \int_0^t i(t) dt$$

o Energy

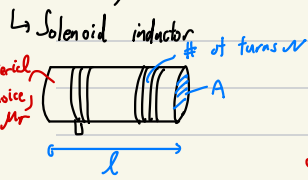
$$w(t) = \frac{1}{2} C v^2(t) = \int_{-\infty}^t p(t) dt = \frac{Q^2}{2C}$$

o Open ckt as $t \rightarrow \infty$

o $v_c(t)$ cannot change abruptly.

↳ $i_c(t) \neq \infty$

Inductor, inductance



o B, Magnetic Flux density [Tesla, T]

$$B_{\text{inside}} = \frac{\mu_r \mu_0 N}{l} i, B_{\text{outside}} = 0$$

↳ μ_r = relative permeability

↳ μ_0 = permeability of free space $= 4\pi \times 10^{-7} \text{ [Tm/A]}$

o Magnetic Flux, $\Phi_m = (B)A = \frac{\mu_r \mu_0 N^2 A}{l} i$

o Inductance, $L = \frac{\Phi_m}{i} = \frac{\mu_r \mu_0 N^2 A}{l} \left[H = \frac{Wb}{A} = \frac{Vs}{A} \right]$

o Voltage

$$v(t) = L \frac{di(t)}{dt}$$

o Power

$$p(t) = v(t) i(t)$$

o Current

$$i(t) = i_o + \frac{1}{L} \int_0^t v(t) dt$$

o Energy

$$w(t) = \int_{-\infty}^t p(t) dt = \frac{1}{2} L i(t)^2$$

o Short ckt as $t \rightarrow \infty$

o $i_L(t)$ cannot change abruptly.

↳ $v_L(t) \neq \infty$

Equivalent Capacitance

o Parallel:

$$C_{eq} = C_1 + C_2 + C_3 \dots$$

$$Q_1 = C_1 V, Q_2 = C_2 V, Q_3 = C_3 V \dots$$

o Series:

$$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \dots \right)^{-1}$$

$$v_1 = \frac{Q}{C_1}, v_2 = \frac{Q}{C_2}, v_3 = \frac{Q}{C_3} \dots$$

Equivalent Inductance

o Parallel:

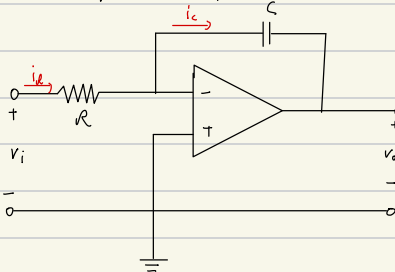
$$L_{eq} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} \dots \right)^{-1}$$

o Series:

$$L_{eq} = L_1 + L_2 + L_3$$

o Integrator

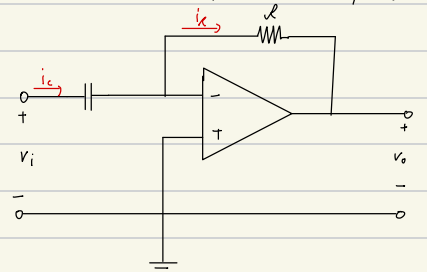
↳ An op amp circuit whose output is proportional to the integral of the input signal



$$v_o = -\frac{1}{RC} \int_0^t v_i(t) dt$$

o Differentiator

↳ An op amp circuit whose output is proportional to the rate of change (derivative) of input signal



$$v_o = -RC \frac{dv_i}{dt}$$